

- Wraps a single literal integer. Its `interpret()` is the **base case** of the recursion: it just returns its own value, with no further calls.
- These are the **leaves** of the AST — the recursion bottoms out here.

AddExpression / SubtractExpression — the non-terminal rules

```
public class AddExpression implements IExpression {
    private IExpression leftExpression;
    private IExpression rightExpression;

    public AddExpression(IExpression leftExpression, IExpression rightExpression) {
        this.leftExpression = leftExpression;
        this.rightExpression = rightExpression;
    }

    @Override
    public int interpret() {
        return leftExpression.interpret() + rightExpression.interpret();
    }
}
```

- Each holds **two child IExpression s** (typed as the interface, so a child can be a number *or* another whole subtree).
- **interpret() is recursive:** it evaluates its left child, evaluates its right child, and combines them (+ here, - in SubtractExpression). This is the **composite recursion** that walks the tree — an internal node's value is defined in terms of its children's values.
- SubtractExpression is identical except it returns `left - right`.

Context — turning a string into a tree (the parser)

```
public IExpression parseExpression(String expression) {
    String[] tokens = expression.split(" ");
    IExpression result = new NumberExpression(Integer.parseInt(tokens[0]));
    for (int i = 1; i < tokens.length - 1; i += 2) {
        String operator = tokens[i];
        IExpression right = new NumberExpression(Integer.parseInt(tokens[i + 1]));
        if (operator.equals("+")) {
            result = new AddExpression(result, right);
        } else if (operator.equals("-")) {
            result = new SubtractExpression(result, right);
        }
    }
    return result;
}
```

- `split(" ")` breaks `"5 + 3 - 2"` into tokens `["5", "+", "3", "-", "2"]`. (This tiny "lexer" assumes single-space separation.)
- **result starts as the first number** — `NumberExpression(5)`.
- **The loop walks the rest in (operator, number) pairs** (`i += 2`): it reads an operator and the number after it, then **wraps the running result** as the left child and the new number as the right child. So the tree grows left-deep:
- after `+ 3`: `result = Add( Number(5), Number(3) )`
- after `- 2`: `result = Subtract( Add(Number(5),Number(3)), Number(2) )`
- `i < tokens.length - 1` stops one short so `tokens[i + 1]` is always a valid operand — it never reads past the end.
- The returned `result` is the **root** of the AST. Nothing has been evaluated yet — parsing only *builds* the tree; `interpret()` evaluates it.

InterpreterDesignPattern — the demo

```
String inputExpression = "5 + 3 - 2";

Context context = new Context();
IExpression expression = context.parseExpression(inputExpression); // build the tree
int result = expression.interpret(); // evaluate the tree

System.out.println("Result: " + result); // 6
```

- **Two distinct phases:** first `parseExpression` builds the AST, then `interpret()` evaluates it. This separation is the essence of the pattern — represent the sentence as objects, then run one operation over that representation.
- `interpret()` on the root triggers the recursive cascade down to the leaves, producing `6`.

Why the design decisions

Why represent the expression as a tree of objects at all?

Because it turns "evaluate this string" into "walk this structure," and the structure mirrors the grammar exactly. Each grammar rule is one class; evaluating is just polymorphic recursion. Once the AST exists you can also do *other* things with it later (pretty-print it, optimize it, translate it) by adding new operations — the representation is reusable, a plain `eval(string)` function is not.

Why split into terminal vs. non-terminal expressions?

That split **is** the grammar. A number is a terminal (it stands alone); an addition is a non-terminal (it's defined using other expressions). Modeling both as `IExpression` lets them nest arbitrarily — `AddExpression` doesn't know or care whether its children are literals or deep sub-trees, because they're all just `IExpression`s. That uniformity is what makes the recursion clean.

Why is `interpret()` recursive, and why does the interface make that possible?

A non-terminal's meaning depends on its parts, so evaluation is naturally recursive: to add, first evaluate the two operands (which may themselves be additions/subtractions). Typing the children as `IExpression` means each node calls `interpret()` on its children **without a type check** — the terminal case (`NumberExpression`) ends the recursion by returning a value directly. This is the same idea as the Composite pattern applied to grammar rules.

Why separate parsing (`Context`) from interpreting (`interpret()`)?

Two responsibilities: **building** the tree from text vs. **evaluating** the tree. Keeping them apart means the expression classes know nothing about string syntax (they'd work if you built the tree by hand, `new AddExpression(new NumberExpression(5), ...)`), and the parser knows nothing about evaluation. You can change the input syntax without touching evaluation, and vice versa.

Why does this build the tree left-to-right (and what was wrong before)?

For `+` and `-` — which have **equal precedence and are left-associative** — left-to-right nesting is exactly correct: `5 + 3 - 2` must mean `(5 + 3) - 2`, and the loop produces precisely that left-deep tree.

Bug that was fixed. The original `Context` used a **stack that popped two operands per operator** — that's a *postfix / Reverse-Polish* evaluator, e.g. for `"5 3 + 2 -"`. But the demo input is **infix** (`"5 + 3 - 2"`). On the first operator the stack had only one operand, so the second `pop()` threw `java.util.EmptyStackException` and the program crashed. The parser was rewritten to build the tree left-to-right from the infix input, so `"5 + 3 - 2"` now evaluates to `6`.

Note this left-to-right approach is correct **only** because the language has just `+`/`-`. If you add `*` or `/`, it would ignore precedence (it'd compute `2 + 3 * 4` as `20`, not `14`) — you'd then need operator-precedence handling (e.g. the shunting-yard algorithm) or a recursive-descent parser.

Execution flow (the demo — `"5 + 3 - 2"`)

Phase 1 — parse (build the AST):

```
tokens = [ "5", "+", "3", "-", "2" ]

result = Number(5)
i=1: "+" 3 → result = Add( Number(5), Number(3) )
i=3: "-" 2 → result = Subtract( Add(Number(5),Number(3)), Number(2) )

returns the root:
```

```
graph TD
    Subtract --> Add
    Subtract --> N2[Number(2)]
    Add --> N5[Number(5)]
    Add --> N3[Number(3)]
```

Phase 2 — interpret (evaluate the AST):

```
Subtract.interpret()
├── left = Add.interpret()
│   ├── left = Number(5).interpret() → 5
│   └── right = Number(3).interpret() → 3
│       └── returns 5 + 3 = 8
```

```
└─ right = Number(2).interpret() → 2
└─ returns 8 - 2 = 6
```

Console output:

```
Interpreter Design Pattern
Evaluating Expression: 5 + 3 - 2
Result: 6
```

Notes / possible extensions (not changed in the code)

- **Adding a new operation is easy** — e.g. `MultiplyExpression` implements `IExpression` with `left * right`, plus a `*` branch in the parser. That's the pattern's sweet spot: one new class per new grammar rule. (Precedence, as noted, is the extra work.)
- **The parser is deliberately minimal** — it assumes well-formed, single-space-separated input and doesn't handle parentheses or validation. Interpreter is about *evaluating the tree*, not about industrial-strength parsing; real languages use a dedicated parser/lexer to build the AST and keep Interpreter for the evaluation step.
- **Plain main, no Spring** — the pattern is pure OO structure and needs no container.

Relationship to the other Behavioral patterns

- **Interpreter (this module)** — model a grammar as a class-per-rule tree and evaluate it recursively via `interpret()`.
- **Composite** (structural) — Interpreter's AST *is* a Composite; the terminal/non-terminal split is leaf vs. composite node.
- **Visitor** — often paired with Interpreter to add *many* operations over the AST (evaluate, print, type-check) without editing the expression classes.
- **Strategy / State** — delegate to a single swapped-in object; Interpreter instead walks a whole structure of objects.

Reach for Interpreter when you have a **small, stable grammar** you need to evaluate repeatedly (arithmetic, boolean rules, query filters, config mini-languages) and modeling each rule as a class keeps the evaluation clean and extensible. For large or fast-changing languages, a real parser generator is the better tool.